



Analytics Solutions Bootcamp: #06

Data Governance Essentials

An introduction to governing data through ownership, quality, metadata, and access controls.

Hello, everyone! Welcome to the sixth module of the Analytics Solutions Bootcamp. Today, you will learn about the essentials of data governance.

Topics to be covered



- What is data governance?
- Ownership and stewardship
- Metadata, glossary, catalog, and lineage
- Data quality governance
- Master and reference data
- Classification, privacy, and access

What is data governance?

Understand how data governance establishes trust, accountability, and control over organizational data

What is data governance?

How many active customers? >>

Someone asks: *"How many active customers do we have?"*

- Marketing dashboard: **17,542** active customers
- Finance dashboard: **15,821** active customers
- Operations dashboard: **18,093** active customers

Which dashboard is correct?

Ask the audience what their thoughts are

What is data governance?

Why don't the numbers match? >>

Several governance problems could produce different answers.

- Definition: What does "active customer" mean?
- Source: Which customer dataset should be used?
- Quality: Are there duplicates or missing records?
- Ownership: Who decides which definition is correct?
- Accountability: Who resolves problems when they occur?

What is data governance?

Data governance defined



Data Governance is defined as the exercise of authority and control (planning, monitoring, and enforcement) over the management of data assets through policies, standards, roles, responsibilities, and processes.

It helps organizations answer questions such as (1) who owns the data? (2) what does the data mean? (3) which data should we trust? (4) who can access the data? (5) what happens when something goes wrong?

What is data governance?

Data management vs governance >>

Data management focuses on activities performed throughout the data lifecycle.

Data governance establishes the authority, accountability, standards, and controls surrounding those activities

Data Management	Data Governance
Store customer data	Define who owns customer data
Build data pipelines	Establish data quality expectations
Transform data	Define standard business terms
Provide data access	Determine who should have access
Delete/archive data	Establish retention requirements

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What is data governance?

Elements of data governance >>

A practical data governance program brings together:

- **People:** ownership, stewardship, accountability
- **Processes:** how governance activities are performed
- **Policies & standards:** rules for managing and using data
- **Technology:** tools that help implement and monitor governance

Data governance is not just a technology problem.

Data ownership and stewardship

Understand who is accountable for defining, managing, and using organizational data

Data domains



Organizations can group related data into data domains representing major areas of the business.

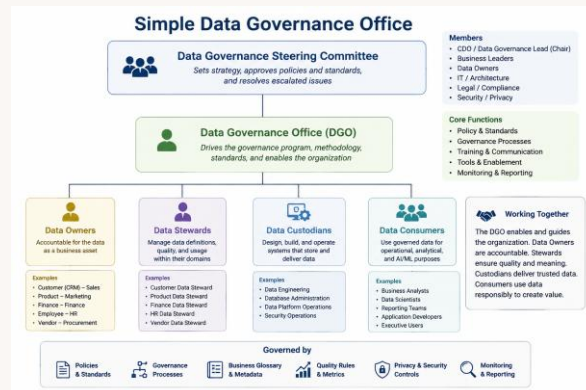
- **Customer:** Customer profile, contact information, customer status
- **Product:** Product name, category, pricing
- **Employee:** Employee profile, position, department
- **Finance:** Revenue, expenses, accounts
- **Vendor:** Supplier information, contracts

Whenever we are assigning data ownership, we can start from identifying what data domains an organization has.

Data ownership and stewardship

Data governance office (DGO) >>

A data governance office hierarchy typically follows a three-to-five-tier structure that moves from executive oversight down to daily operational execution.



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Whenever we are assigning data ownership, we can start from identifying what data domains an organization has.

Who is responsible for data? >>

- **Data Owner:** Accountable for a data domain and decisions surrounding its use.
- **Data Steward:** Maintains definitions, standards, rules, and quality of the data.
- **Data Custodian:** Technically stores, protects, and maintains the data.
- **Data Consumer:** Uses the data for operational or analytical purposes.

Data ownership and stewardship

Example: Employee data



Key idea: Ownership does not necessarily mean physically storing or processing the data.

Role	Example	Responsibility
Data Owner	HR Director	Accountable for employee data
Data Steward	HR Operations	Maintains definitions and quality
Data Custodian	IT / Data Engineering	Stores and protects data
Data Consumer	HR Analyst	Uses data for analysis

Data ownership and stewardship

Governance through RACI



RACI framework can clarify responsibilities for governance activities.

- R – Responsible: performs the work
- A – Accountable: ultimately owns the outcome
- C – Consulted: provides input
- I – Informed: needs to know the outcome

Data ownership and stewardship

Example of RACI framework

>>

Activity	HR	Steward	Data Engineer	Analyst
Define active employee	A	R	C	C
Implement rule	C	C	R	I
Monitor quality	A	R	C	C
Build HR dashboard	C	C	C	R

Data ownership and stewardship

Activity: Who owns the problem? >>

Scenario: A dashboard shows 1,250 employees, while payroll reports 1,218. Who should:

Investigate the discrepancy?

Define what counts as an active employee?

Correct the pipeline if there is a technical issue?

Approve the official definition?

Who should change the dashboard from 1,250 to 1,218?

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- Investigate the discrepancy? Data Steward (HR Operations)
- Define what counts as an active employee? Data Steward (HR Operations)
- Correct the pipeline if there is a technical issue? Data Custodian (Data Engineer)
- Approve the official definition? Data Owner (HR Director)
- Who should change the dashboard from 1,250 to 1,218?
 - The investigation might reveal that:
 - Payroll excludes employees on unpaid leave.
 - The HR system includes them as active employees.
 - The dashboard uses the HR definition.
 - Both numbers are technically correct **for different definitions**.
 - So the actual problem may not be **bad data**. It could be **inconsistent business definitions**.

Metadata, glossary, catalog, and lineage

Document, discover, understand, and trace
organizational data

Timing: 30 seconds

Metadata: data about data



Metadata includes information about technical and business processes, data rules and constraints, and logical and physical data structures.

- Business metadata: domains, entities, metrics
- Technical metadata: table, column, type, and transformation
- Operational metadata: refresh time, status, usage, and incidents
- Governance connects all three to accountable people and controlled processes.

We will focus on business metadata. One of the examples of business metadata is business glossary.

Business glossary



A **glossary** is a means of sharing this vocabulary or **business term** within the organization. Developing and documenting standard data definitions reduces ambiguity and improves communication. **Definitions must be clear**, rigorous in wording, and explain any exceptions, synonyms or variants

Business term	Business category	Definition	Business term status	Business owner	Origin
Applicant individual	Grants	A person (not representing an organization) who has applied for a grant from the agency, which has not yet been approved, or was unsuccessful.	Approved	Director, Grants Management	Grants Management System
Signed grant agreement	Grants	A grant agreement that has been signed by the grantee and the agency under seal and uploaded to the agency's recordkeeping system.	Under review	Director, Grants Management	General Records Authority 28
Online visit	Publishing	Unique external use of our website comprising at least one session in a defined date range.	Draft	Director, Communications	Google Analytics
Onsite visitor	Security	People who are issued a visitor's pass to enter an agency building, who are not staff or contractors.	Approved	Director, Facilities	Agency Protective Security Policy
Date of onsite visit	Security	The date an onsite visitor was issued a visitor's pass (note: passes are issued for one day only).	Approved	Director, Facilities	Agency Protective Security Policy

Example: <https://www.naa.gov.au/information-management/describing-information/metadata/managing-metadata/developing-business-glossary>

Data dictionary



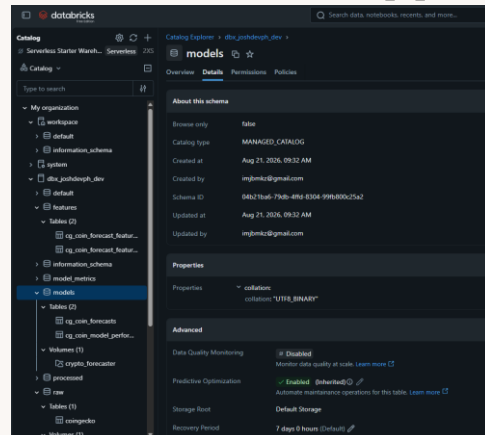
A **data dictionary** defines the structure and contents of data sets, often for a single database, application, or warehouse. The dictionary can be used to manage the names, descriptions, structure, characteristics, storage requirements, default values, relationships, uniqueness, and other attributes of every data element in a model.

Field Name	Data type	Field Length	Constraint	Description
Client_id	Int	10	Primary key	Client id, Auto generated
Client_name	Varchar	20	Not null	Name of client
Password	Varchar2	30	Not null	Login Password for client
Contact_no	Int	15	Not null	Landline or mobile number
Email_id	Varchar2	30	Not null	Any email id
Max_Users	Int	10	Not null	Maximum number of users
Current_users	Int	10	Not null	Currently present user

Metadata, glossary, catalog, and lineage

Data catalog

Searchable inventory of data assets with metadata, schemas, ownership, lineage, and quality metrics, enabling discovery and governance through documentation



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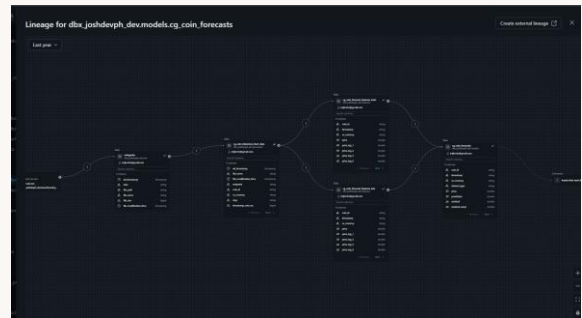
<https://www.databricks.com/blog/what-is-data-catalog>

Metadata, glossary, catalog, and lineage

Data lineage



Data lineage is the process of recording, tracking and visualizing data and AI over time, from origin to consumption. Effective data lineage provides data teams with an end-to-end view of how data is transformed and flows across their data estate.



<https://www.databricks.com/blog/what-is-data-lineage>

Metadata, glossary, catalog, and lineage

Goals of creating these assets >>

Users should be able to find:

- what data exists and where it is located
- definitions, owners, classifications, and quality status
- access instructions and approved uses

<https://www.databricks.com/blog/what-is-data-lineage>

Data quality governance

Turn quality expectations into owned, measurable controls

Timing: 30 seconds

A review on data quality >>

- Data quality measures how well a dataset meets criteria.
- Common **dimensions** of data quality.
 - Accuracy, completeness, consistency, integrity, reasonability, timeliness, uniqueness, and validity.

How do we measure the quality of our data? We have dimensions of data quality.

Data quality dimensions



Data Quality Dimension	Definition	Example
Accuracy	Data correctly represents the real-world object or event.	Customer birthdate stored as 1985-03-15 when the actual birthdate is 1995-03-15.
Completeness	All required data is present.	Customer record is missing email and phone number. "Missing" may mean something else.
Consistency	Data is the same across systems and datasets.	Ex. 1: Customer status is Active in CRM but Inactive in billing system. Ex. 2: On customer gender, there are MALE, FEMALE, M, F, 0, 1
Integrity	Relationships between data elements are valid and maintained.	A sales transaction contains Customer ID 99999, which does not exist in the customer table.

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Data Quality Dimension

Definition

Good Example

Bad Example

Accuracy

- Data correctly represents the real-world object or event.
- Customer birthdate is **1995-03-15**, matching official records.
- Customer birthdate stored as **1985-03-15** when the actual birthdate is 1995-03-15.

Completeness

- All required data is present.
- Customer record contains name, email, phone number, and address.
- Customer record is missing email and phone number.

Consistency

- Data is the same across systems and datasets.
- Customer status is **Active** in both CRM and billing systems.

- Customer status is **Active** in CRM but **Inactive** in billing system.

Integrity

- Relationships between data elements are valid and maintained.
- Every sales transaction references a valid Customer ID.
- A sales transaction contains Customer ID **99999**, which does not exist in the customer table.

Data quality dimensions



Data Quality Dimension	Definition	Example
Reasonability	Data values make logical and business sense.	Employee age is 250 years old.
Timeliness	Data is available and up-to-date when needed.	Inventory was last updated three months ago.
Uniqueness	Each real-world entity is represented only once.	Three customer records exist for the same customer with slightly different spellings.
Validity	Data conforms to defined business rules and formats.	Email stored as user@email or 12345.

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Data Quality Dimension

Definition

Good Example

Bad Example

Reasonability

- Data values make logical and business sense.
- Employee age is **35 years old**.
- Employee age is **250 years old**.

Timeliness

- Data is available and up-to-date when needed.
- Inventory levels are updated every hour.
- Inventory was last updated three months ago.

Uniqueness

- Each real-world entity is represented only once.
- One customer record exists for Juan Dela Cruz.
- Three customer records exist for the same customer with slightly different

spellings.

Validity

- Data conforms to defined business rules and formats.
- Email stored as user@email.com.
- Email stored as **user@email** or **12345**.

Data quality governance



Data governance teams are responsible for the following.

- Define data quality rules
- Monitor data quality
- Resolve data quality issues

Role	Responsibility for data quality rules
Data Owner	Accountable: approves quality expectations, thresholds, and important business rules
Data Steward	Responsible: defines, documents, maintains, and monitors data quality rules
Data Custodian	Implements: translates rules into technical checks, pipelines, alerts, etc.
Data Consumer	Consulted: provides requirements and reports quality issues encountered during use

Data quality governance

Operationalizing data quality



- Define data quality rules
- Apply and measure data quality
- Monitor data quality
- Detect and prioritize
- Resolve data quality issues
- Validate and improve



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Data quality scorecard



- Implement the applicable data quality rules for each column in a dataset.
- Flag records that did not meet data quality expectations.
- Aggregate the data quality findings to set score for each column and the entire dataset.
- Set thresholds on data quality scores.

Data quality governance

Data quality scorecard



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Master and reference data

Create consistent identities and controlled shared values

Timing: 30 seconds

Define master and reference data>>

Master Data is data about the business entities (e.g., employees, customers, products, financial structures, assets, and locations) that provide context for business transactions and analysis.

Reference Data is any data used to characterize or classify other data, or to relate data to information external to an organization. The most basic Reference Data consists of codes, descriptions, mappings, and hierarchies (e.g. status, location codes)

Reference data management >>

Reference Data Management entails control and maintenance of defined domain values, definitions, and the relationships within and across domain values.

- Reference data structures: lists, cross-reference lists, taxonomies, ontologies
- Types of reference data: proprietary or internal, industry, geographic, computational
- Maintain standard reference data set metadata

Master data management



Master Data Management entails control over Master Data values and identifiers that enable consistent use, across systems, of the most accurate and timely data about essential business entities.

- System of record and system of reference
- Trusted source, golden record
- Entity resolution

Entity resolution



Entity resolution is the process of determining whether two references to real world objects refer to the same object or to different objects

Source ID	Name	Address	Telephone
123	John Smith	123 Main, Dataland, SQ 98765	
234	J. Smith	123 Main, Dataland, DA	2345678900
345	Jane Smith	123 Main, Dataland, DA	234-567-8900

Master and reference data – Entity resolution

Validate, standardize, enrich >>

Data has to be validated. Standardize based on standard formatting of contents. Enrich by filling in missing details

Source ID	Name	Address	Telephone
123	John Smith	123 Main, Dataland, SQ 98765	
234	J. Smith	123 Main, Dataland, SQ 98765	+1 234 567 8900
345	Jane Smith	123 Main, Dataland, SQ 98765	+1 234 567 8900

Matching and identity resolution >>

How do we match records? We need to establish our rules when

1. Similarity matching (fuzzy match, Levenshtein, soundex)
2. Identify matching rules based on other dimensions (full name, date of birth, tax number, phone number, email address)
3. Define hierarchy of rules and handle missing records

1. Similarity matching on short names, Chinese names, common names,
2. Which dimensions to prioritize?
3. What if relevant details are missing?

Matching and identity resolution >>

- If two records share the same last name, first name, birth date, and SSN, but the street address differs, is it safe to assume they refer to the same person who has changed their mailing address?
- If two records share the same SSN, street address, and first name, but the last name differs, is it safe to assume they refer to the same person who has changed their last name? Would the likelihood be increased or decreased based on gender and age?
- How do these examples change if the SSN is unknown for one record? What other identifiers are useful to determine the likelihood of a match? How much confidence is required for the organization to assert a match?

Master and reference data – Entity resolution

Matching and identity resolution >>

Source ID	Name	Address	Telephone	Candidate ID	Party ID
123	John Smith	123 Main, Dataland, SQ 98765		XYZ	1
234	J. Smith	123 Main, Dataland, SQ 98765	+1 234 567 8900	XYZ, ABC	2
345	Jane Smith	123 Main, Dataland, SQ 98765	+1 234 567 8900	ABC	2

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Conclusions on entity resolution >>

- Despite the best efforts, match decisions sometimes prove to be incorrect.
- Maintain the history of matches so that matches can be undone when discovered to be incorrect.
- Match rate metrics enable organizations to monitor the impact and effectiveness of their matching inference rules.
- Reprocessing of match rules can help identify better match candidates as new information is received by the entity resolution process.

Conclusions on entity resolution >>

- Entity resolution would be a lot simpler on high quality data. High quality data must be enforced throughout all the stages in the customer journey.

Classification, privacy, and access

Apply practical controls according to sensitivity and purpose

Timing: 30 seconds

Classification, privacy, and access

Personal data classifications >>

RA 10173 – Data Privacy Act of 2012

- **Personal Information:** Information that identifies or can identify an individual.
- **Sensitive Personal Information:** Personal information requiring greater protection due to its nature (e.g. health, education, government-issued information).
- **Privileged Information:** Information considered privileged communication under applicable laws.

Classification, privacy, and access

Organizational data classifications>>

- **Public:** Approved for public access and distribution.
- **Internal:** Intended for use within the organization and not for general public distribution.
- **Confidential:** Sensitive information accessible only to authorized individuals with a legitimate business need.
- **Restricted:** Highly sensitive information requiring the strongest access and security controls.

Classification, privacy, and access

Managing PIs and SPIs



Ask: “What would be the use cases of the downstream users for sensitive information?”

- User access matrix (UAM) must be defined and signed-off on
 - Define access control list (RBAC or ABAC)
 - Column limits, row-level security, data masking
 - Pseudonymize (hashing with salt)
 - Anonymize (generalization, categorical, binning, truncating IP, rounding)

Classification, privacy, and access

Reminders and considerations >>

Always ask: “What are my or my users’ rationale when using sensitive information? Would the reporting and analytics be sensible and ethical?”

References

- Association, D. M., Henderson, D., & Earley, S. (2017b). DAMA-DMBOK: Data Management Body of Knowledge.